

```

1  % maximum number of independent zeros
2
3  n = input("Enter n: ");
4
5  A = input(sprintf("Enter %dx%d matrix: ",n,n));
6
7
8
9  grid = zeros(n,n);
10
11 for i=1:n
12     for j=1:n
13         if A(i,j) == 0
14             grid(i,j) = 1;
15         end
16     end
17 end
18
19 [answer, points] = find_max_indep(A, grid, n);
20
21 disp("Max number of independent zeros: ")
22 disp(answer)
23 disp("Max corresponding indices: ")
24 disp(points(1:end-1,:))
25
26 if answer < n
27     disp("Number of independent zeros are less than n")
28
29     for i=1:n
30         min_row = min(A(i,:));
31         A(i,:) = A(i,:) - min_row;
32     end
33
34     for j=1:n
35         min_col = min(A(:,j));
36         A(:,j) = A(:,j) - min_col;
37     end
38
39     A
40
41     [answer, points] = find_max_indep(A, grid, n);
42
43     if answer == n
44         disp("Now, matrix has n zeros")
45     else
46         disp("Matrix still does not have n zeros")
47         disp("Lets find the min number of lines to cross all zeros")
48
49         [r_i, c_i] = my_helper(A,n);
50
51         roww = zeros(n,1);
52         coll = zeros(n,1);
53
54         for i=1:size(r_i,2)
55             roww(r_i(i), 1) = 1;
56         end
57
58         for i=1:size(c_i,2)

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59         coll(c_i(i), 1) = 1;
60     end
61
62     minmin = min(min(A(r_i, c_i)));
63
64     A(r_i, c_i) = A(r_i, c_i) - minmin;
65
66     for i=1:n
67         for j=1:n
68             if roww(i)==0 && coll(i)==0
69                 A(i,j) = A(i,j) + minmin;
70             end
71         end
72     end
73
74     A
75
76
77 end
78 end
79 %%
80
81 function [my_answer, my_points] = find_max_indep(A, grid, n)
82     my_answer = 0;
83     [r,c] = size(A);
84
85     one_present = 0;
86
87     my_points = [0 0];
88
89     for i=1:n
90         for j=1:n
91             if grid(i,j) == 1
92                 one_present = 1;
93                 break;
94             end
95         end
96     end
97
98     if one_present == 0
99         return
100     end
101
102     my_answer_temp = 0;
103
104     % final_i = -1;
105     % final_j = -1;
106     % flag = 0;
107
108     for i=1:r
109         for j=1:c
110             if grid(i,j) == 1
111                 % flag = 1;
112                 grid_cpy = grid;
113                 grid_cpy(i,:) = 0;
114                 grid_cpy(:,j) = 0;
115                 % new_mtx = [A(1:i-1,1:j-1) A(1:i-1,j+1:end); A(i+1:end, 1:j-1)
A(i+1:end, j+1:end)];

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116         % my_answer_temp = max(my_answer_temp, 1+max(my_answer,
find_max_indep(A(i+1:end, j+1:end))));
117         [this_ans, points] = find_max_indep(A, grid_cpy,n);
118         new_var = 1+max(my_answer, this_ans);
119         if new_var > my_answer_temp
120             % my_answer_temp = new_var
121             my_points = [i j; points];
122             my_answer_temp = max(my_answer_temp, new_var);
123             % final_j = j;
124             % final_i = i;
125         end
126     end
127 end
128 end
129
130 % if flag == 1
131 %     disp([final_i, final_j])
132 % end
133
134 my_answer = max(my_answer, my_answer_temp);
135 end
136
137 %%
138 function [r_i, c_i] = my_helper(A,n)
139     for i=1:n
140         for j=1:n
141             one_n = 1:n;
142             rows_included = nchoosek(one_n, n-i);
143             cols_included = nchoosek(one_n, n-j);
144
145             rr = size(rows_included, 1);
146             cc = size(cols_included, 1);
147
148             for aa = 1:rr
149                 for bb = 1:cc
150                     min_mtx = min(min(A(rows_included(aa,:), cols_included(bb,:))));
151
152                     if min_mtx > 0
153                         r_i = rows_included(aa,:);
154                         c_i = cols_included(bb,:);
155                         disp("Rows not crossed: ")
156                         disp(r_i)
157                         disp("Columns not crossed: ")
158                         disp(c_i)
159                         return
160                     end
161                 end
162             end
163         end
164     end
165 end
166
167 end

```